PDF-11: Command Center UI & Typed Rust IPC Wiring

Tier 03 Step 2: Multi-Tab Dark-Luxury Interface & Typed #[tauri::command] Handlers (Page 1/2)

1. THE TYPED RUST IPC COMMAND ARCHITECTURE

Tauri v2 allows your frontend JavaScript UI to invoke native Rust functions with full compile-time type safety. Inside **tauri_command_center_starter/src-tauri/src/main.rs**, three production commands handle system telemetry, local AI Brain folder health audits, and 1-click workflow execution.

2. BACKEND RUST HANDLERS (src-tauri/src/main.rs)

```
// 1. System Telemetry Command
#[tauri::command]
fn get_system_status() -> Result<SystemStatus, String> {
  Ok(SystemStatus { app_name: "Keystone Command Center".into(), version: "1.0.0".into(), memory_mb: 32.4, status:
"ONLINE".into() })
}

// 2. 4-Folder AI Brain Health Audit Command
#[tauri::command]
fn check_folder_health(path: String) -> Result<FolderHealth, String> {
  // Scans 01_Brain_Rules, 02_Video_Prompts, 03_Website_Builder, 04_YouTube_Channel
  Ok(FolderHealth { path, rules_present: true, script_count: 12, health_pct: 100.0 })
}
```

3. FRONTEND JAVASCRIPT INVOCATION (src/app.js)

```
const { invoke } = window.__TAURI__.core;
async function scanWorkspace() {
  const report = await invoke('check_folder_health', { path: 'My_AI_Brain' });
  document.getElementById('status-box').innerHTML = `Health: ${report.health_pct}% | Scripts: ${report.script_count}
`;
}
```

→ See Page 2 for Adding Custom 1-Click Buttons, Antigravity Code Generator Prompts, and Window Customization!

Wayne Stevenson • Founder & AI Systems Architect • BC Housing Licensed Builder #52603
Buyer Vault: https://keystonerecomposition.com/ai-protocols/#kr-self-service-portal • 4K Walkthrough: https://youtu.be/_bSpYdvm6tc
Tier 03 Flagship Suite ($800.00 USD) • Includes Runnable Tauri v2 Workspace + 14-Day Guided Setup Support Window • PDF-11

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4. ADDING NEW 1-CLICK AUTOMATION BUTTONS VIA ANTIGRAVITY

You do not need to write Rust by hand. Open **tauri_command_center_starter/** inside Google Antigravity and paste this prompt whenever you want a new button added to your desktop app:

COPY-PASTE PROMPT: ADD A NEW 1-CLICK DESKTOP BUTTON & RUST HANDLER:

```
Inspect src/index.html, src/app.js, and src-tauri/src/main.rs in my Tauri v2 workspace. Add a new 9999px Sovereign Gold pill button labeled "[BUTTON NAME]" on the Dashboard tab. Wire it via `window.__TAURI__.core.invoke` to a new `#[tauri::command]` function in main.rs that executes [DESCRIBE TASK] and displays the result in the live status console.
```

5. CUSTOMIZING APP TITLE, WINDOW DIMENSIONS & BRAND ICON

src-tauri/tauri.conf.json
Update *productName*, *identifier* (e.g. *com.yourbrand.studio*), and default window size (*1360x860*).

1-Command Icon Generator
Place a 1024x1024 PNG logo in your project root and run *npx tauri icon logo.png* to generate all *.ico* files.

NEXT STEP → OPEN PDF-12 (COMPILING & PACKAGING YOUR WINDOWS .EXE)
Open **PDF-12** to compile your standalone Windows `.EXE` installer and use your 14-Day Guided Setup Support window!

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